

Secant Method

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1 Introduction

The Secant Method is a numerical root-finding algorithm used to find a solution of the nonlinear equation

$$f(x) = 0.$$

Like the False Position Method, it approximates the root by computing the x-intercept of the secant line passing through two points on the function. However, unlike the False Position Method, the Secant Method does not require the initial approximations to bracket the root. Instead, it always uses the two most recent approximations to construct the next secant line.

2 Mathematical Foundation

2.1 Secant Line

Suppose two points $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$. The equation of the secant line passing through these two points is

$$y = f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x - x_{n-1}).$$

2.2 Secant Method

The next approximation is obtained by computing the x-intercept of the secant line. Since the x-intercept satisfies $y = 0$, substituting this condition into the secant-line equation gives

$$0 = f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x_{n+1} - x_{n-1}).$$

Solving this equation for x_{n+1} gives:

$$x_{n+1} = x_{n-1} - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}f(x_{n-1}).$$

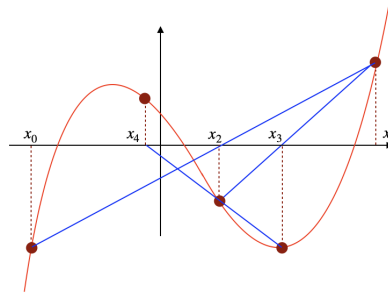


Figure 1: Secant Method

3 Algorithm Flow

At every iteration, a secant line is constructed using the two most recent approximations. The x-intercept of the secant line is taken as the next approximation to the root. After computing the new approximation, the older approximation is discarded, and the two most recent approximations are retained for the next iteration. The process continues until the convergence criterion is satisfied or the maximum number of iterations is reached.

Algorithm 1 Secant Method

Input: Function $f(x)$, Initial approximations x_0 and x_1 , Tolerance ϵ , Maximum iterations N

Output: Approximate root x

```
for  $k = 1$  to  $N$  do
  if  $|f(x_1) - f(x_0)| < \delta$  then
    return failure
  end if
   $x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0)$ .
  if  $|f(x_2)| < \epsilon$  then
    return  $x_2$ 
  end if
   $x_0 = x_1$ 
   $x_1 = x_2$ 
end for
return failure
```

4 Convergence Analysis

4.1 Order of Convergence

The order of convergence is a measure of how rapidly an iterative method approaches the exact solution. Consider a sequence x_n that converges to r . The sequence is said to converge with order α if there exists a positive constant λ such that

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - r|}{|x_n - r|^\alpha} = \lambda,$$

where $\lambda > 0$ is called the asymptotic error constant. In numerical methods, let

$$e_n = x_n - r$$

denote the error at the n -th iteration. The above definition can then be written as

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \lambda.$$

For sufficiently large n ,

$$|e_{n+1}| \approx \lambda |e_n|^\alpha.$$

The value of α determines the speed of convergence. Larger values of α correspond to faster convergence. In general,

- $\alpha = 1$: Linear convergence
- $1 < \alpha < 2$: Superlinear convergence
- $\alpha = 2$: Quadratic convergence

4.2 Error equation of the Secant Method

The Secant iteration is

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n).$$

Let r be the exact root of $f(x) = 0$, and define the error $e_n = x_n - r$. Since $x_n = r + e_n$, we have

$$f(x_n) = f(r + e_n).$$

Taylor expansion gives

$$f(r + e_n) = f(r) + f'(r)e_n + \frac{1}{2}f''(r)e_n^2 + \dots$$

Since $f(r) = 0$, we obtain

$$f(x_n) = f'(r)e_n + \frac{1}{2}f''(r)e_n^2 + \dots$$

Similarly,

$$f(x_{n-1}) = f'(r)e_{n-1} + \frac{1}{2}f''(r)e_{n-1}^2 + \dots$$

The denominator is $f(x_n) - f(x_{n-1})$. Substitute the Taylor expansions:

$$f(x_n) - f(x_{n-1}) = f'(r)(e_n - e_{n-1}) + \frac{1}{2}f''(r)(e_n^2 - e_{n-1}^2) + \dots$$

We can write

$$f(x_n) - f(x_{n-1}) = (f'(r) + \frac{1}{2}f''(r)(e_n + e_{n-1}) + \dots)(e_n - e_{n-1})$$

The numerator is $f(x_n)(x_n - x_{n-1})$. Since $x_n - x_{n-1} = e_n - e_{n-1}$, we have

$$f(x_n)(x_n - x_{n-1}) = (f'(r)e_n + \frac{1}{2}f''(r)e_n^2 + \dots)(e_n - e_{n-1}).$$

Since both the numerator and denominator contain the common factor $e_n - e_{n-1}$, it follows that

$$e_{n+1} = e_n - \frac{f'(r)e_n + \frac{1}{2}f''(r)e_n^2 + \dots}{f'(r) + \frac{1}{2}f''(r)(e_n + e_{n-1}) + \dots}.$$

Since e_n and e_{n-1} are sufficiently small near the root, we neglect the higher-order terms and expand the fraction in powers of the errors. After simplification, we have

$$e_{n+1} = \frac{f''(r)}{2f'(r)} e_n e_{n-1} + O(e^3).$$

Finally, define $C = \frac{f''(r)}{2f'(r)}$ so that $e_{n+1} \approx C e_n e_{n-1}$.

Assuming the method converges with order p , i.e., $e_{n+1} \approx Ae_n^p$. By shifting the iteration index by one, we obtain, $e_n \approx Ae_{n-1}^p$. Solving for e_{n-1} gives,

$$e_{n-1} \approx \left(\frac{e_n}{A}\right)^{1/p}.$$

Substituting this expression into $e_{n+1} \approx Ce_n e_{n-1}$, we have

$$e_{n+1} \approx \frac{C}{A^{1/p}} e_n^{(1+1/p)}.$$

But we also assumed $e_{n+1} \approx Ae_n^p$. So we now have two expressions for the same quantity. Since these describe the same asymptotic behavior as $e_n \rightarrow 0$, the exponents of e_n must be equal. Therefore, $p = 1 + \frac{1}{p}$, or equivalently, $p^2 = p + 1$. Solving the quadratic equation gives

$$p = \frac{1 + \sqrt{5}}{2} \approx 1.6180.$$

Therefore, the Secant Method converges superlinearly with order $p = \frac{1+\sqrt{5}}{2} \approx 1.6180$, which is faster than the Bisection and False Position methods (linear convergence), while requiring only function evaluations. Compared with Newton's Method, it avoids computing derivatives at the cost of a slightly slower convergence rate.

5 Numerical Example

To illustrate the Secant Method, consider the nonlinear equation

$$f(x) = x^2 - 2.$$

The exact root is

$$x = \sqrt{2} \approx 1.41421356.$$

Choose the initial points

$$x_0 = 1, x_1 = 2.$$

Applying the Secant Method produces the following sequence of approximations.

Table 1: $f(x) = x^2 - 2$		
Iteration	Approximation	$f(x)$
1	1.3333	-0.2222
2	1.4	-0.04
3	1.41463	0.00119
4	1.41421	-6.00729×10^{-6}
5	1.41421	-8.93146×10^{-10}

After 5 iterations, the approximate root is 1.41421.